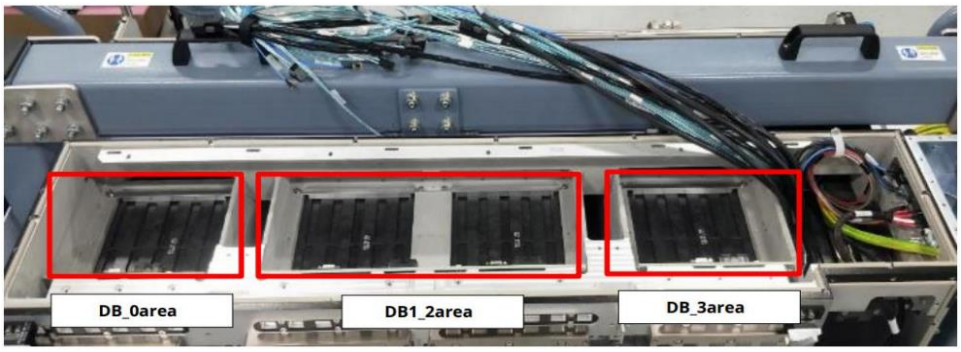
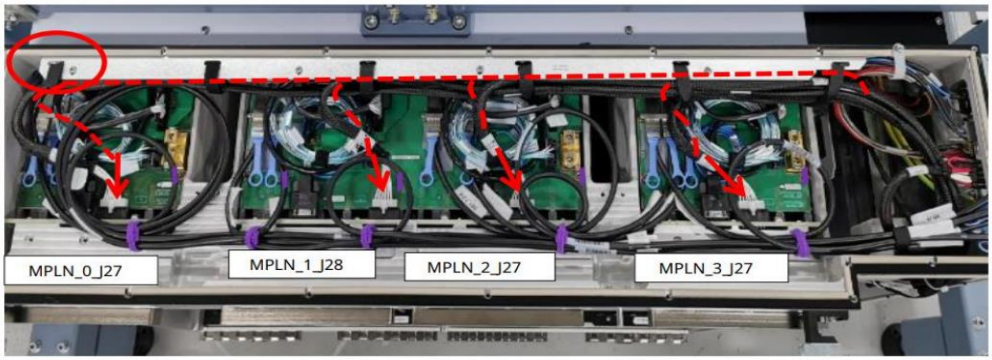
Build Overview:

1 **Distribution Board Install**
2 **Standoff Install**
3 **Route and Install Cables**
4 **Install the following cables**
   a.QSFP
   b.Midplane Power





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

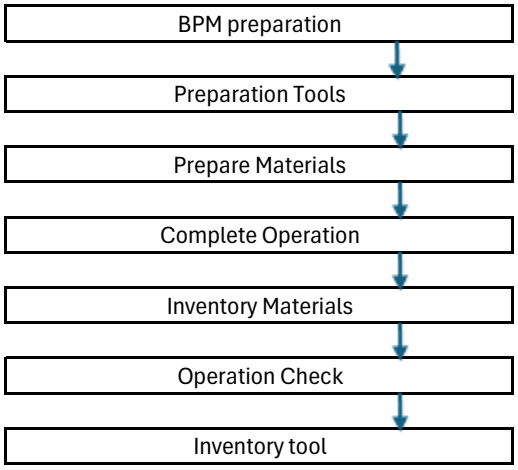
MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	813-929-00	HDW VELCRO STRP 5/8	7
2	630-926-00	ELEC-MECH, INSTR HLA, UFII DB12 DISTRIBUTION BOARD, FOR SEMI TEST	1
3	630-925-01	ELEC-MECH, INSTR HLA, UFII DB6 DISTRIBUTION BOARD, FOR SEMI TEST	2
4	651-093-00	PLASTIC DB6/12 STANDOFF W/ HOLDER OTHER ARTICLES	4
5	470-483-04	SCRW CP BUT HD M5X.	4
6	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER. O.D. 11MM	5
7	938-195-01	CABLE TIE MOUNT 1/4	5
8	626-643-00	CABLE QSFP ISL CU TELECOM WITH CONN	2
9	626-643-01	CABLE QSFP1 ISL CU TELECOM WITH CONN	1
10	626-944-00	CABLE QSFP PCIE SR CU TELECOM WITH CONN	1
11	626-644-01	CABLE QSFP1 PCIE SR CU TELECOM WITH CONN	1
12	626-644-02	CABLE QSFP2 PCIE SR CU TELECOM WITH CONN	1
13	626-644-03	CABLE QSFP3 PCIE SR CU TELECOM WITH CONN	1
14	360-174-01	VELCRO, 10PCS 1/4" X 5" VELCRO BRAND ONE-WRAP TIES 10 CT-91594, PURPLE	5
15	627-093-00	CABLE UF2 MIDPLANE POWER TELECOM WITH CONN	1
16	627-093-01	CABLE UF2 MIDPLANE POWER TELECOM WITH CONN	1
17	813-929-00	HDW VELCRO STRP 5/8	4

METHOD:

Flow chart of operation steps



MACHINE:

TOOL NAME	TORQUE VALUE
FLAT BLADE SCREWDRIVER	0.19 N.m
PH2 PHILIPS TORQUE SCREWDRIVER	1.6 N.m
3MM HEXAGON SOCKET HEAD	0.7 N.m
DB TOOL	NA

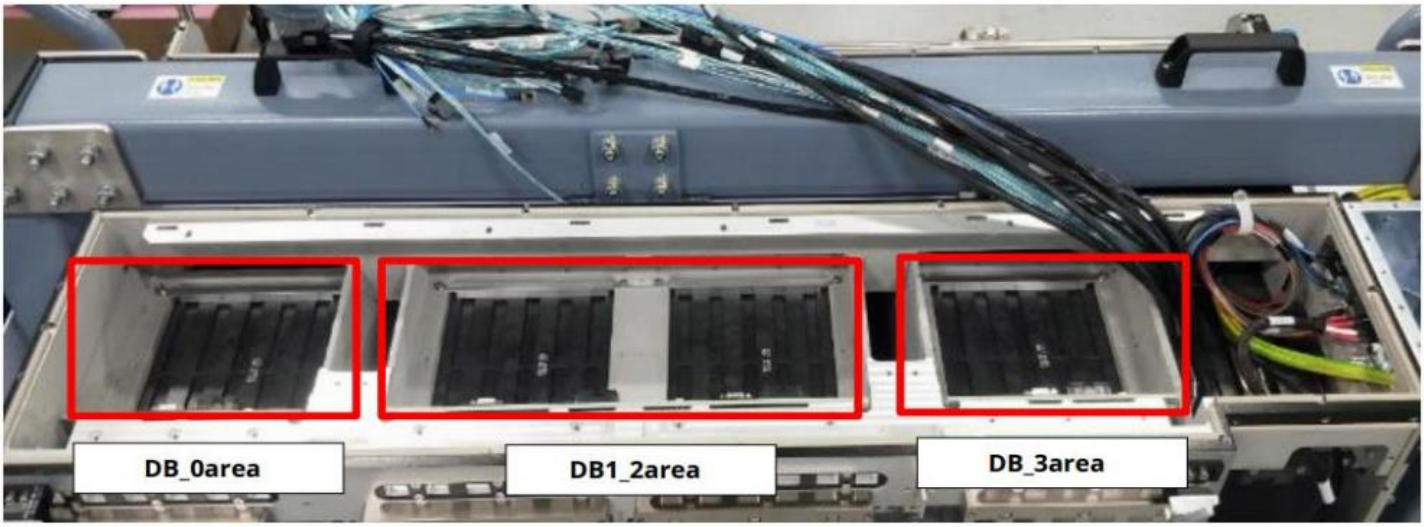
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD: Operation steps

1.1 Turn T/H into direction as shown in the figure, and tie the previously installed wires to the outer frame of the cart, leaving enough space to install the power distribution board. as the picture shows
Use Velcro strap (813-929-00) to tie up the cable for temporary



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	630-926-00	ELEC-MECH, INSTR HLA, UFII DB1B DISTRIBUTION BOARD, FOR SE	1

MACHINE: DB Tool

METHOD:

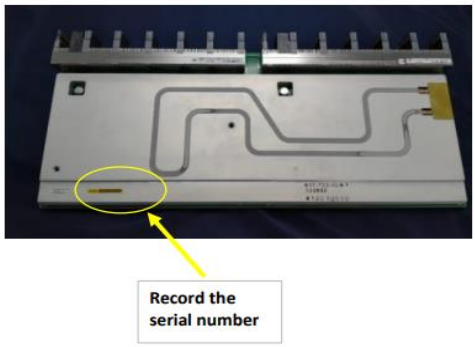
2.1 Install the DB12 power distribution board.

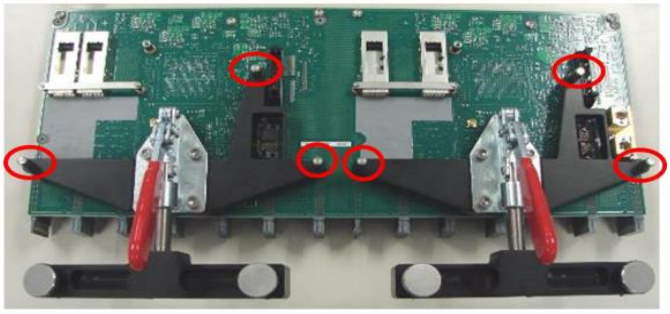
a. Take 1 DB12(630-926-00) and record the serial number.

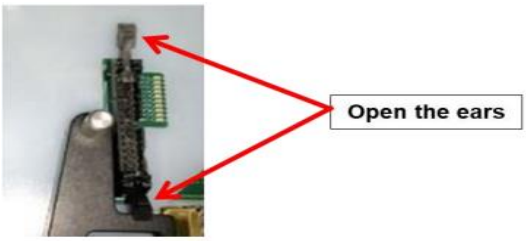
Note: ESD protection measures must be taken when removing and handling the distribution board.

b. Take two designated tools (DB tool) for the power distribution board, use the studs on the 6 clamps to install on the power distribution board as shown in the figure, and manually tighten the thumb screws.

c. Open the upper ears of the ribbon connectors, a total of 4 ears







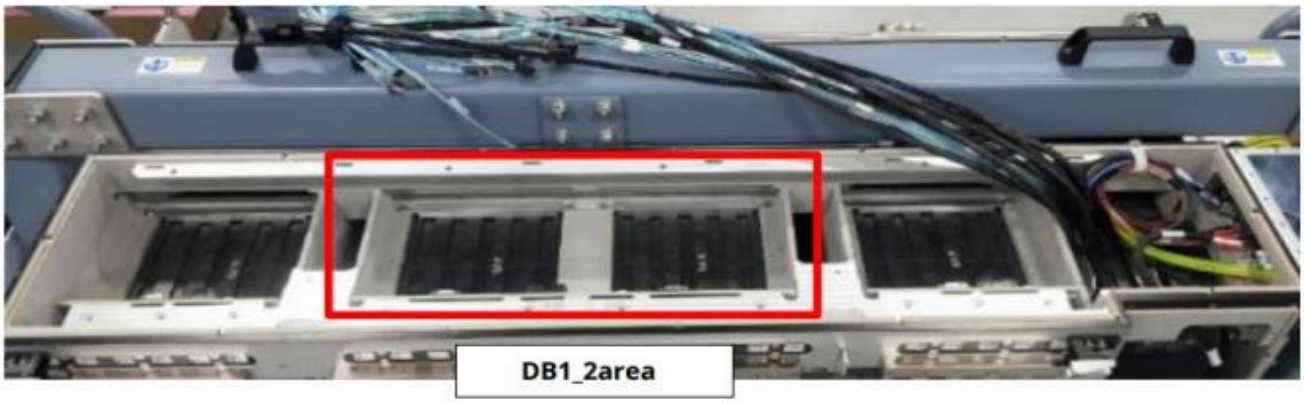
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

d. Install the DB12 power distribution board at the DB1_2 area in the figure below



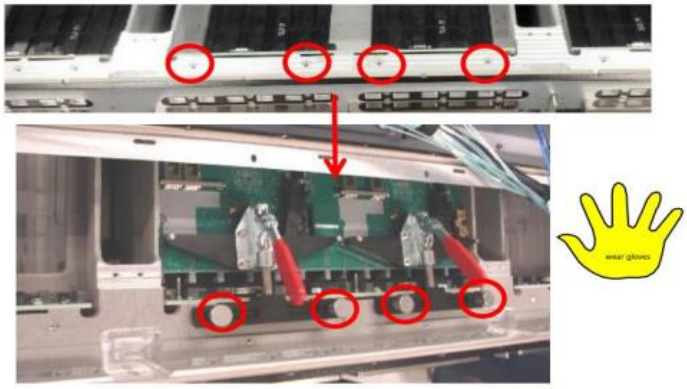
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

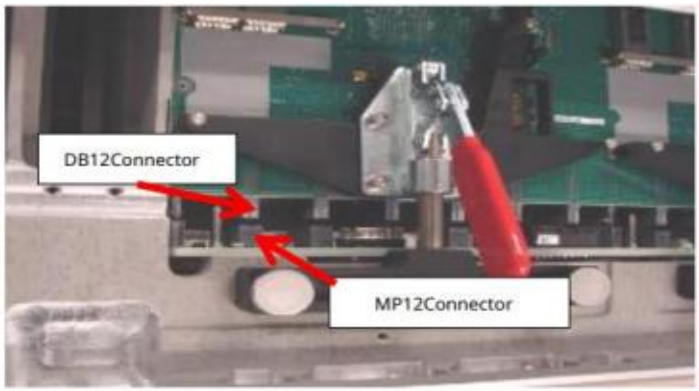
MACHINE: N/A

METHOD:

f. Carefully place the power distribution board with the special tool locked into the T/H frame as shown in the figure, and the screws on the fixture should be consistent with the holes on the T/H frame. Fasten the clamp to the T/H by hand-tightening the screws on the clamp.



g. Align all DB12 connectors with the previously installed MP12 connectors.



MAN: CERTIFIED OPERATOR, TECHNICIAN

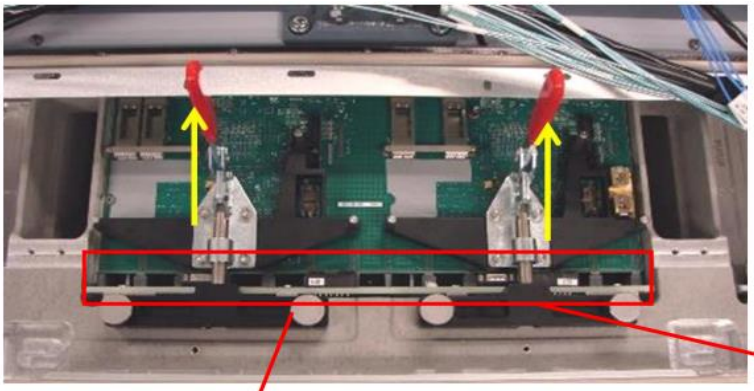
MATERIAL: N/A

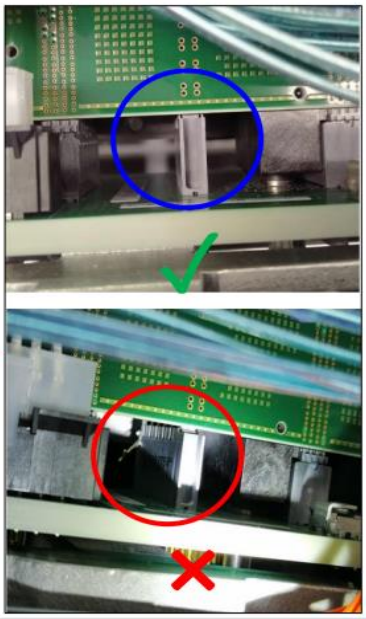
MACHINE: N/A

METHOD:

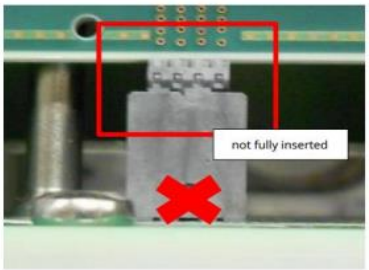
h. As shown in the figure, push the slide bar away from the worker to install the DB12 board.

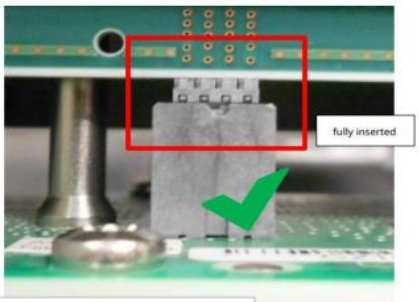
i. Make sure the PCBA board is fully inserted into all the connector as shown.





Check all the connector, make sure no pin bent or damage connector.





Check all the connector, make sure all connector are fully inserted into midplane and no gap.

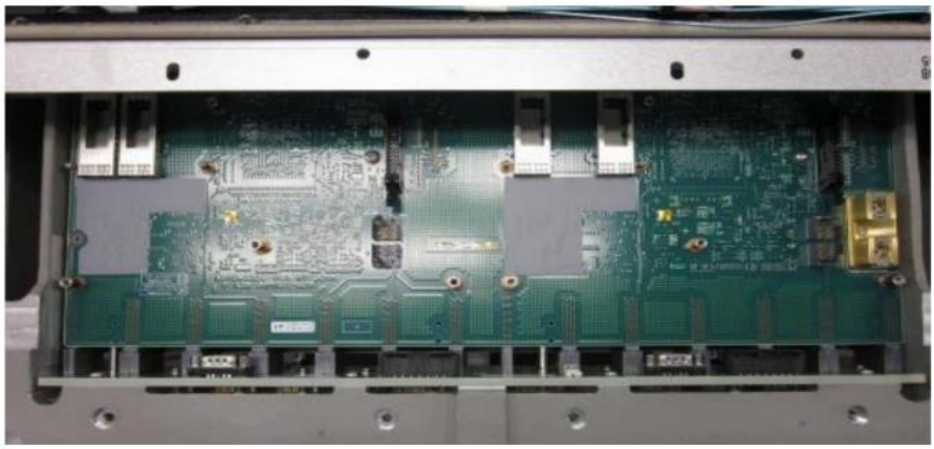
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

j. Remove the studs from the clamps, then remove the large nuts from the clamps, then remove the clamps



InstallDB12behind the board

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
2	630-925-01	ELEC-MECH, INSTR HLA, UFII DB6 DISTRIBUTION BOARD, FOR SE	2

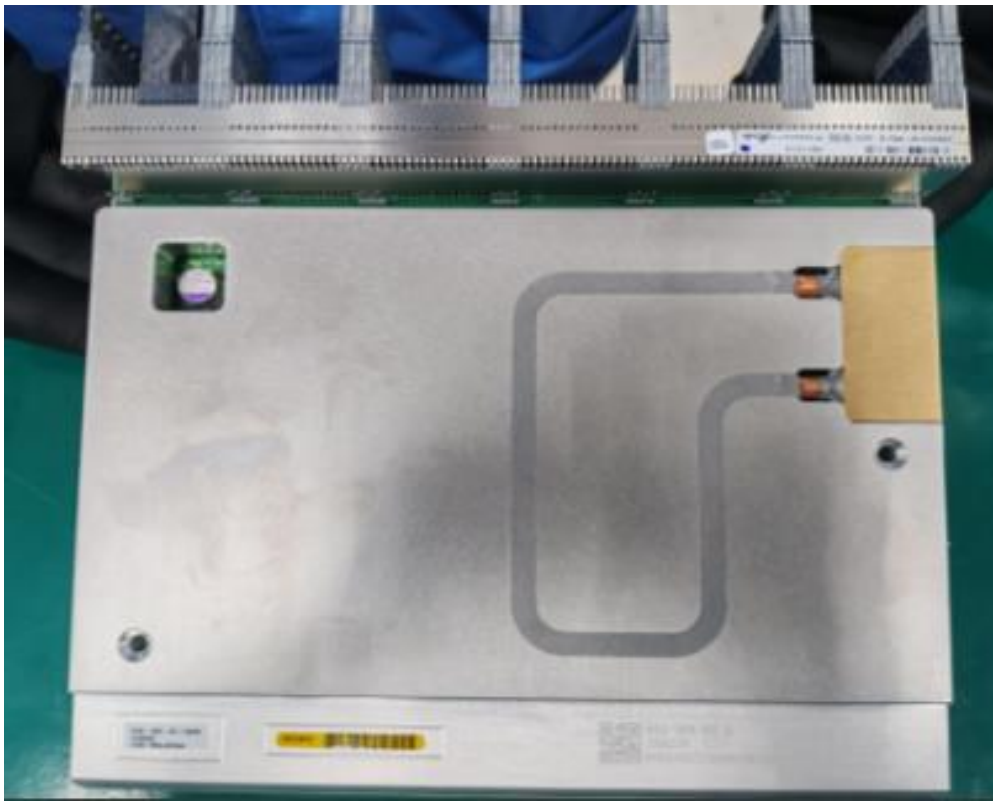
MACHINE: N/A

METHOD:

3.1 Install the DB6 Distribution Board

a. Take 2 DB6 Board (630-925-01), and record the serial number

Note: ESD protection measures must be taken when removing and handling the distribution board.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

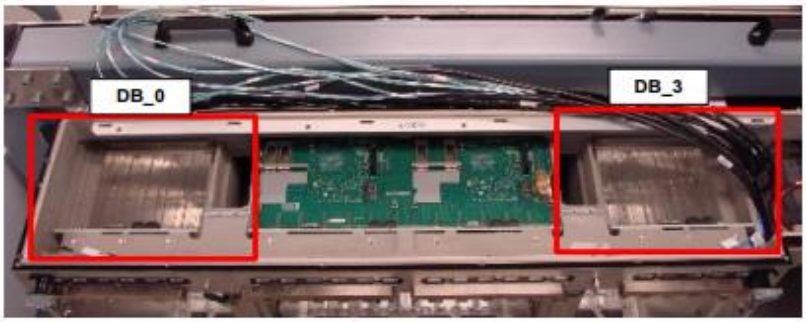
MACHINE: DB Tool

METHOD:

b. Take a designated tools (DB Tool) for the DB6 board, use the studs on the 3 clamps to install on the power distribution board as shown in the figure, and manually tighten the thumb screws.

c. Open the upper ears of the connector, a total of 2 ears.

e. Free out the area of DB_0 and DB_3 to install DB6 board.



MAN: CERTIFIED OPERATOR, TECHNICIAN

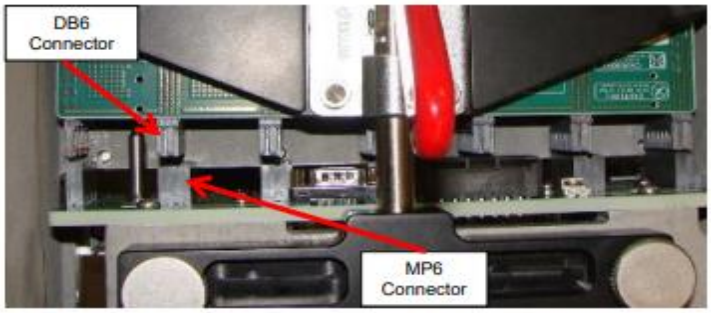
MATERIAL: N/A

MACHINE: N/A

METHOD:

f. Put the power distribution board with the special tool locked into the T/H frame as shown in the figure, and keep the screws on the fixture consistent with the holes on the T/H frame. Fasten the clamp to the T/H by hand-tightening the screws on the clamp.

g. Align the DB6 connector with the previously installed MP6 connector to keep the same position.



MAN: CERTIFIED OPERATOR, TECHNICIAN

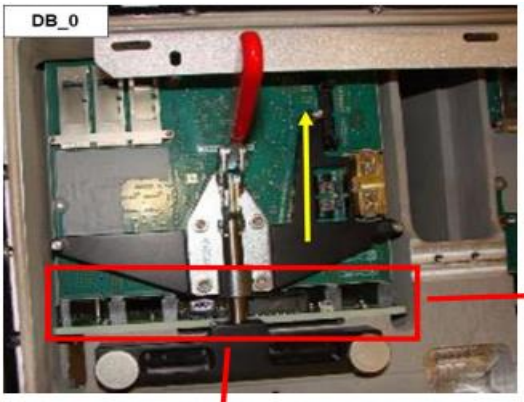
MATERIAL: N/A

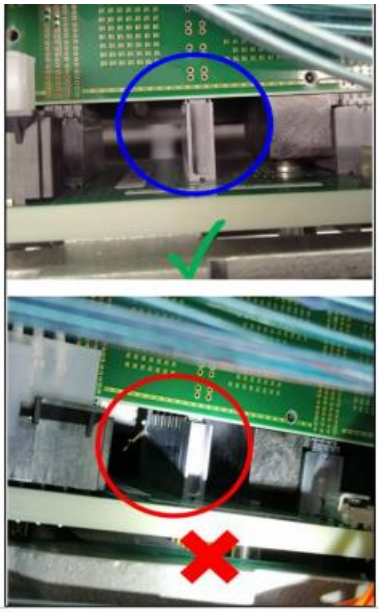
MACHINE: N/A

METHOD:

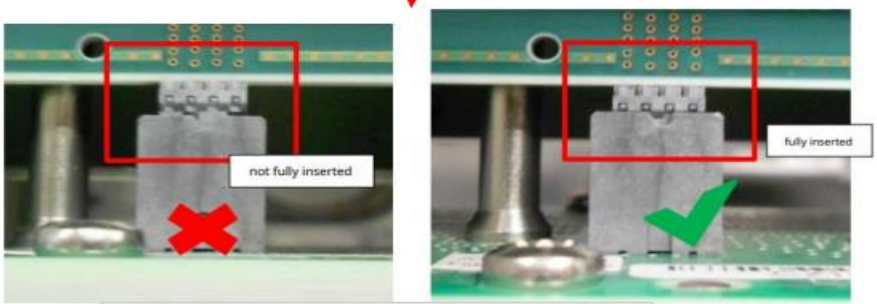
h. As shown in the figure, push the slide bar away from people to install the DB6 board.

i. Make sure the PCBA board is fully inserted into the connector as shown.





Check all the connector, make sure no pin bent or damage connector.



Check all the connector, make sure all connector are fully inserted into midplane and no gap.

MAN: CERTIFIED OPERATOR, TECHNICIAN

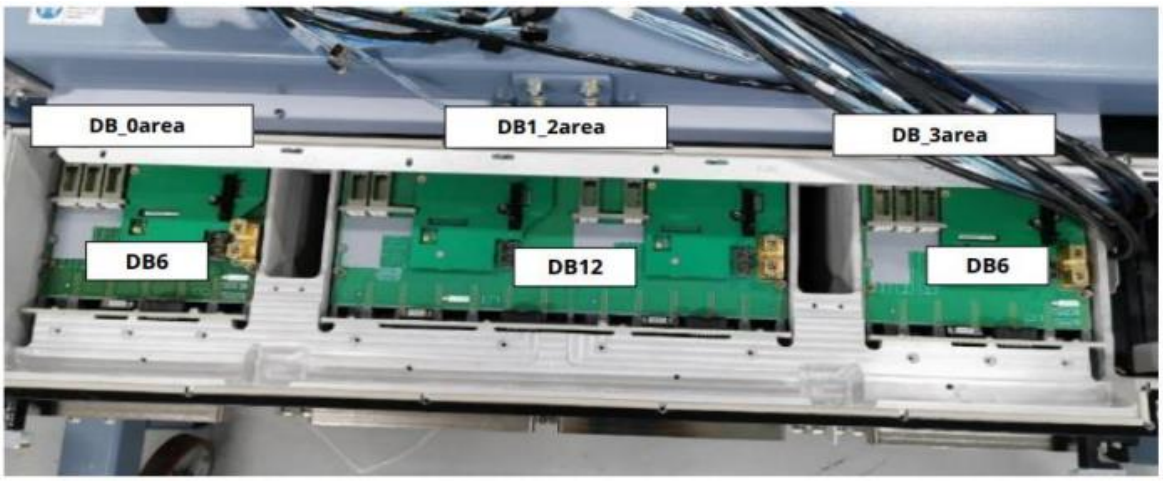
MATERIAL: N/A

MACHINE: N/A

METHOD:

j. Remove the studs from the clamps, then remove the large nuts from the clamps, then remove the clamps.

k. Repeat the above steps to install another DB6 power distribution board in the DB_3 area. As shown in the figure, the schematic diagram after 2 DB6 boards and 1 DB12 board are assembled.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
3	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE: N/A

METHOD:

4.1 DB6 wiring in the right DB 3 area

a. Use a long cable tie (813-929-00) at the right position of the picture to tie all the wires through the slot of the frame aluminum plate.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	813-929-00	HDW VELCRO STRP 5/8	1
5	651-093-00	PLASTIC DB6/DB12 STANDOFF W/ HOLDER OTHER ARTICLES	1

MACHINE:

3mm hexagon bit 0.7 N·m

METHOD:

b. As shown in position 1, use a long cable tie (813-929-00) to temporarily fix all the wires through the slot of the frame. And, one stud will be install between the screw holes of the board and the T/H frame at position 2 as shown in the figure.

c. Take a stud (651-093-00) as shown in the figure, and use a long 3mm ball head screwdriver to fix the stud screw to the screw hole of the power distribution board, and the torque value is 0.7 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

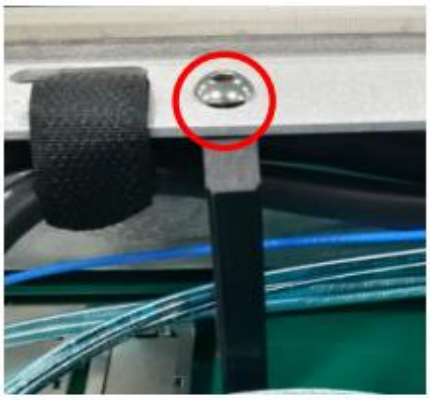
ITEM	PART NUMBER	DESCRIPTION	QTY
6	470-483-04	SCRW CP BUT HD M5X.	1
7	813-929-01		1

MACHINE:

3mm hexagon socket head 0.7 N·m, flat blade 0.19 N·m

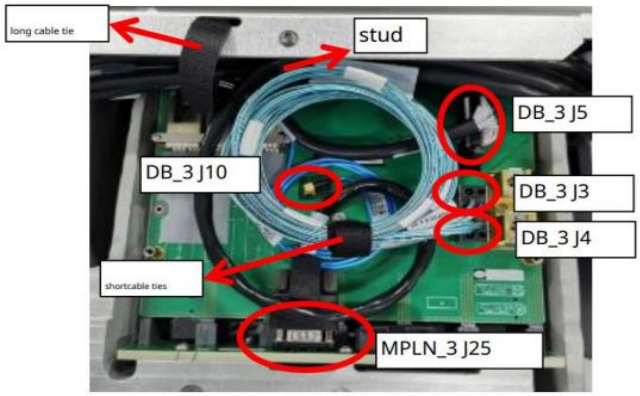
METHOD:

d. Use 1 M5 screw (470-483-04) to fix the top of the stud on the T/H frame. The tool is a 3mm hexagon socket head and the torque value is 0.7 N·m.



e. Find the label in the harness as 'DB_3 J10''DB_3 J3' 'DB_3 J4''DB_3 J5' 'MPLN_3 J25', 'MPLN_3 J25' connector is routed as shown in the figure and fastened to the J25 interface of the MP6 backplane with a flat-blade screwdriver. The torque value is 0.19N·m. The other four wire connectors are inserted into the 4 positions of the power distribution board as shown in the figure according to the line label and the power distribution board connector label.

As shown in the figure, use a short cable tie (813-929-01) to tie the J3 and J4 wires in a circle



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
8	813-929-00	HDW VELCRO STRP 5/8	1
9	651-093-00	PLASTIC DB6/DB12 STANDOFF W/ HOLDER OTHER ARTICLES	1
10	470-483-04	SCRW CP BUT HD M5X.	1
11	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

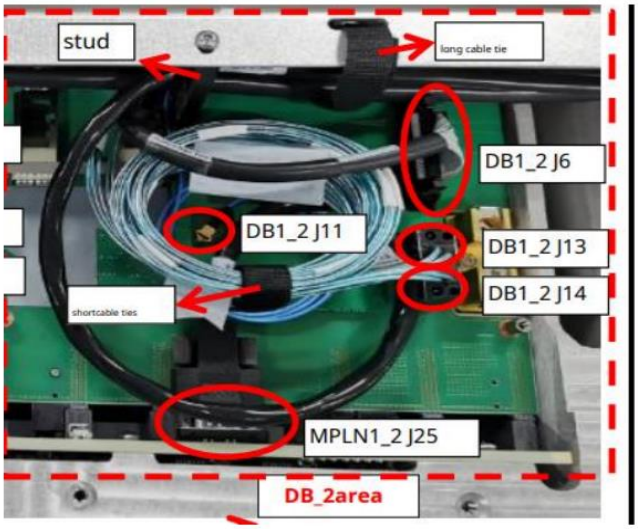
3mm hexagon socket head 0.7 N·m, flat blade 0.19 N·m

METHOD:

5.1 DB12 wiring in the middle DB_2 area
a. Follow step 4.1, use a long cable tie (813-929-00) to temporarily fix all the wires through the slot of the frame, and then install 1 stud (651-093-00) and 1 screw M5 (470-483-04) with 3mm ball head and 0.7 N.m torque.

Find the number in the wire harness for'DB1_2 J13"DB1_2 J14' 'DB1_2 J6"DB1_2 J11"MPLN_1_2 J25' .
The 'MPLN_1_2 J25' connector is routed as shown in the figure and fastened to the J25 interface on the MP12 backplane with a flat-blade screwdriver, with a torque value of 0.19N m

The other four wire connectors are inserted into the The distribution board has 4 positions as shown. As shown in the figure, use a short cable tie (813-929-01) to tie the J13 and J14 wires in a circle.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
12	813-929-00	HDW VELCRO STRP 5/8	1
13	651-093-00	PLASTIC DB6/DB12 STANDOFF W/ HOLDER OTHER ARTICLES	1
14	470-483-04	SCRW CP BUT HD M5X.	1
15	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

3mm hexagon socket head 0.7 N·m, flat blade 0.19 N·m

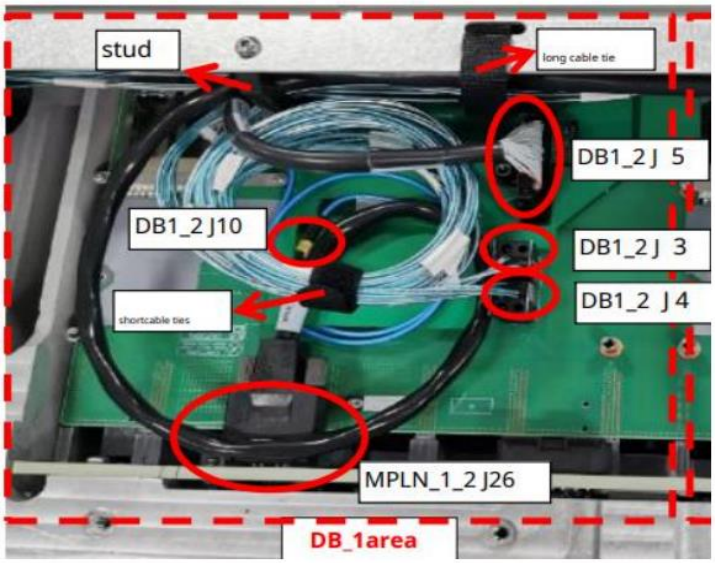
METHOD:

DB12 wiring in the middle DB_1 area

b. Follow step 4.1, use a long cable tie (813-929-00) to temporarily fix all the wires through the slot of the frame, and then install 1 stud (651-093-00) and 1 screw M5 (470-483-04) with 3mm ball head and 0.7 N.m torque.

Find the number in the wire harness for 'DB1_2 J3"DB1_2 J4' 'DB1_2 J5"DB1_2 J10"MPLN_1_2 J26'.
The 'MPLN_1_2 J26' connector is routed as shown in the figure and fastened to the J26 interface on the MP12 backplane with a flat-blade screwdriver, with a torque value of 0.19N m

The other four wire connectors are inserted into the The distribution board has 4 positions as shown. As shown in the figure, use a short cable tie (813-929-01) to tie the J3 and J4 wires in a circle.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
12	813-929-00	HDW VELCRO STRP 5/8	1
13	651-093-00	PLASTIC DB6/DB12 STANDOFF W/ HOLDER OTHER ARTICLES	1
14	470-483-04	SCRW CP BUT HD M5X.	1
15	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

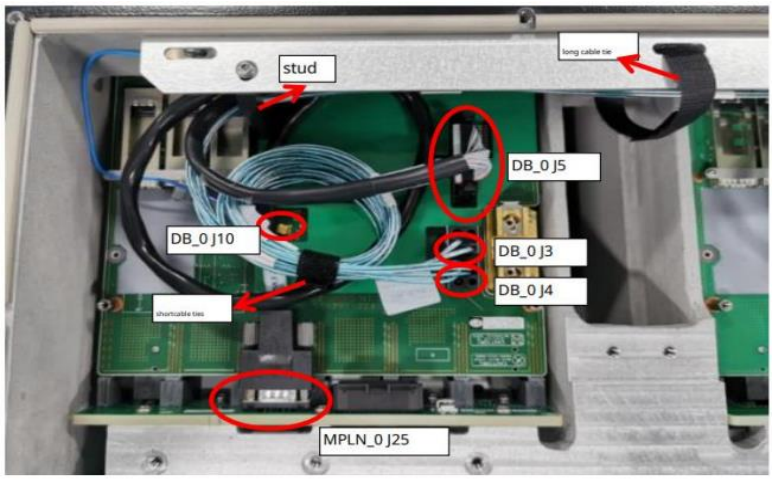
3mm hexagon socket head 0.7 N·m, flat blade 0.19 N·m

METHOD:

6.1 Left DB_0 area DB6 wiring
a. Follow step 4.1, use a long cable tie (813-929-00) to temporarily fix all the wires through the slot of the frame, and then install 1 stud (651-093-00) and 1 screw M5 (470-483-04) with 3mm ball head and 0.7 N.m torque.

Find the number in the wire harness fors'DB_0 J3"DB_0 J4' 'DB_0 J5"DB_0 J10"MPLN_0 J25'.
The 'MPLN_0 J25' connector is routed as shown in the figure and fastened to the J25 interface on the MP12 backplane with a flat-blade screwdriver, with a torque value of 0.19N m

The other four wire connectors are inserted into the The distribution board has 4 positions as shown. As shown in the figure, use a short cable tie (813-929-01) to tie the J3 and J4 wires in a circle.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
20	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/ CONE	5
21	938-195-01	CABLE TIE MOUNT 1/4	5

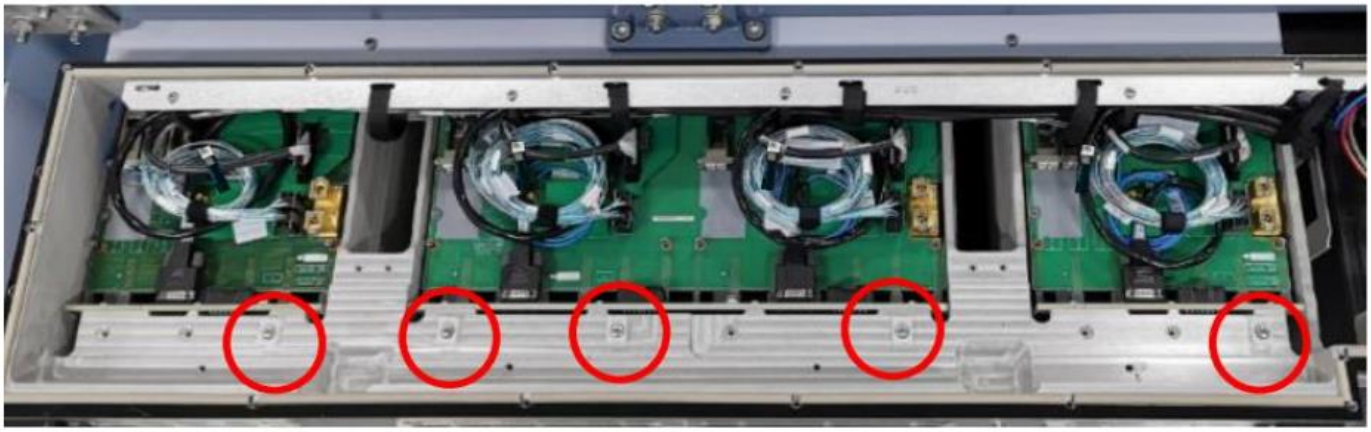
MACHINE:

: PH2 Phillips torque screwdriver 1.6 N·m

METHOD:

7.1 The power distribution board is installed as shown in the figure.

7.2 Use 5 M5 screws (472-426-12) to fix the 5 wire clips (938-195-01) as shown in the figure. The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N·m



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
22	626-643-00	CABLE QSFP ISL CU TELECOM WITH CONN	2
23	626-643-01	CABLE QSFP1 ISL CU TELECOM WITH CONN	1
24	626-644-00	CABLE QSFP PCIE SR CU TELECOM WITH CONN	1

MACHINE:

Marker pen

METHOD:

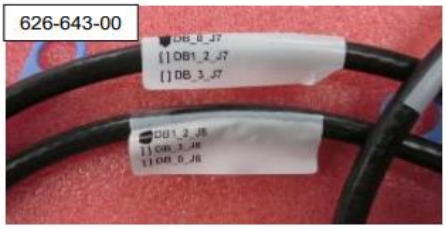
8.1 Take 2 pieces of QSFP ISL cable （626-643-00）, 1 pieces (626-643-01) and 1 piece (626-644-00) cable, and marking label onto the cable.

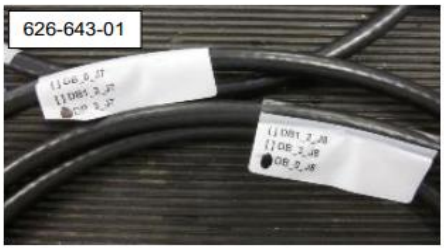
a. Cable （626-643-00） : Label [X] DB_0_J7 and [X] DB1_2_J8.

b. Cable （626-643-00） Label [X] DB1_2_J7 and [X] DB_3_J8.

c.Cable （626-643-01） : Label [X] DB_3_J7 and [X] DB_0_J8.

d. Cable （626-644-00） : Label [X] DB_3_J9 and [X] SB_J7.





MAN: CERTIFIED OPERATOR, TECHNICIAN

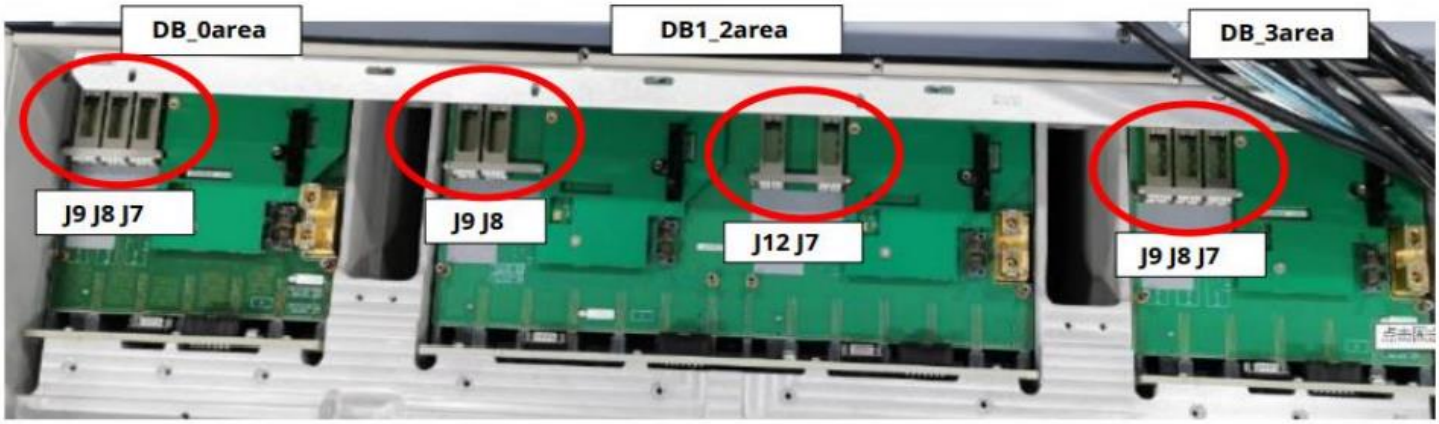
MATERIAL: N/A

MACHINE: N/A

METHOD:

9.1 Cable installation:

a. As shown in the figure, the red circle interface is the wiring location area



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
25	626-644-01	CABLE QSFP1 PCIE SR CU TELECOM WITH CONN	1
26	626-644-02	CABLE QSFP2 PCIE SR CU TELECOM WITH CONN	1
27	626-644-03	CABLE QSFP3 PCIE SR CU TELECOM WITH CONN	1

MACHINE:

N/A

METHOD:

10.1 Install 7 pieces cable 626-643-00(2pcs), 626-643-01 (626-644-00,626-644-01,626-644-02,626-644-03)

a. Install the cable (626-643-00) as shown in the figure, and connect one end to "DB_0_J7" according to the label, and the other end to "DB1_2_J8".

b. Install the cable (626-643-00) as shown in the figure, and connect one end to "DB1_2_J7" according to the label, and the other end to "DB_3_J8".

c. Install the cable (626-643-01) as shown in the figure, and connect one end to "DB_3_J7" according to the label, and the other end to "DB_0_J8".

d. Install the cable (626-644-00) as shown in the picture, and connect one end to "DB_3_J9" and the other end to "SB_J7" according to the label.

e. Install the cable (626-644-01) as shown in the picture, and connect one end to "DB1_2_J9" and the other end to "SB_J5" according to the label.

f. Install the cable (626-644-02) as shown in the figure, and connect one end to "DB_0_J9" and the other end to "SB_J4" according to the label.

g. Install the cable (626-644-03) as shown in the figure, and connect one end to "DB1_2_J12" according to the label, and the other end to "SB_J6".

h. Leave enough length for the connectors 'SB_J4', 'SB_J5', 'SB_J6', and 'SB_J7' as shown in the picture (the end of the connector is just beyond the dog house)

Make sure the end of the connector 'SB_J4', 'SB_J5', 'SB_J6' and 'SB_J7', out of the panel box.

MAN: CERTIFIED OPERATOR, TECHNICIAN

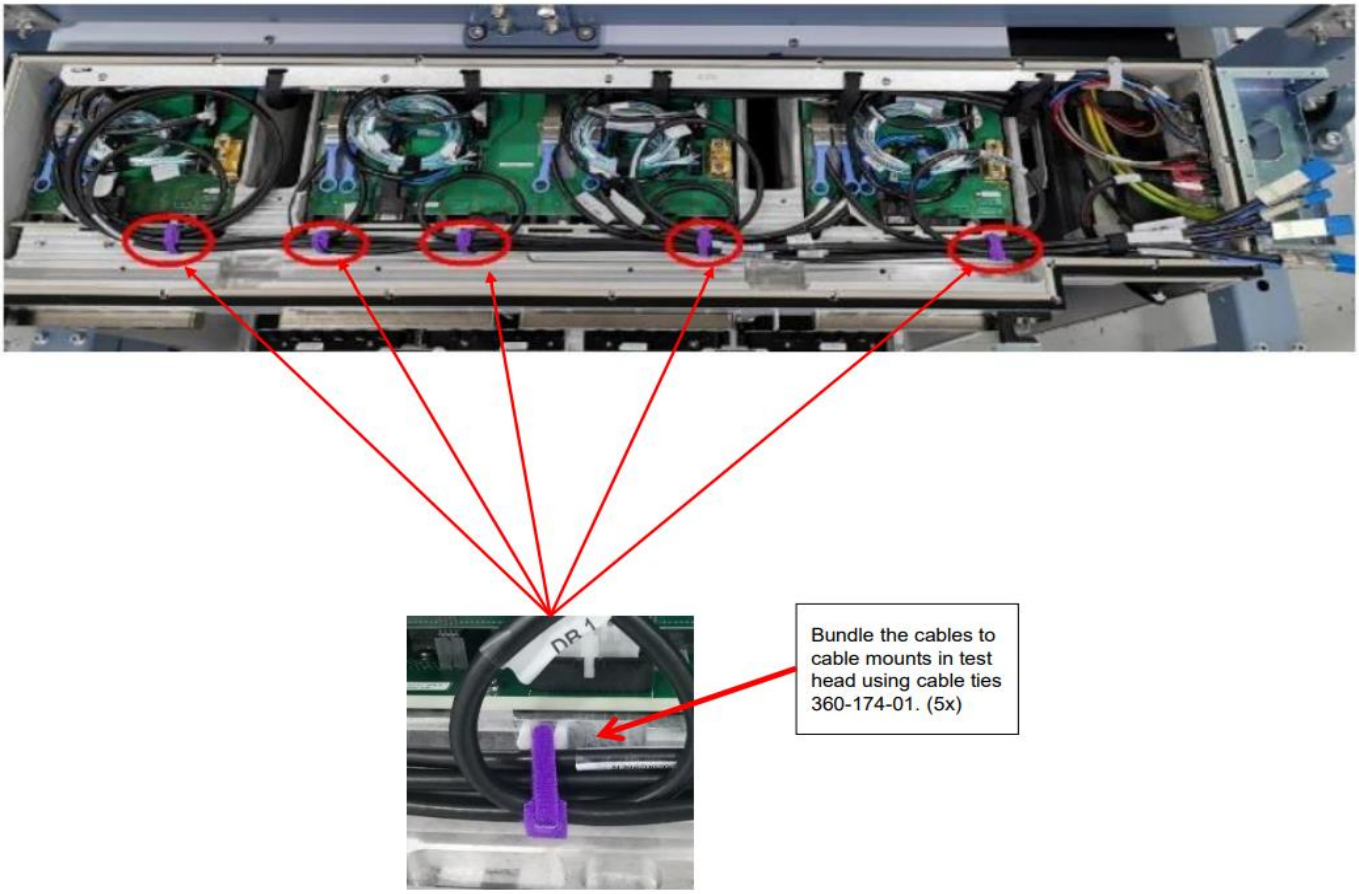
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
28	360-174-01	VELCRO, 10PCS 1/4" X 5" VELCRO BRAND ONE-WRAP TIES 10 CT	5

MACHINE: N/A

METHOD:

11.1 Routing the wires as shown and use the 5 purple cable ties (360-174-01) to tie the 7 wires you just installed.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
29	627-093-00	CABLE UF2 MIDPLANE POWER TELECOM WITH CONN	1
30	627-093-01	CABLE UF2 MIDPLANE POWER TELECOM WITH CONN	1

MACHINE: Marker pen

METHOD:

11.2 Take each 1 piece of Midplane power cable （627-093-XX）， and marking label on the cable.

a. Cable （627-093-00） : Label [X] SP24_J9 and [X] MPLN_0_J27 , [X] MPLN_3_J27.

b. Cable （627-093-01） : Label [X] SP24_J10 and [X] MPLN_1_J28, [X] MPLN_2_J27.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
31	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE: N/A

METHOD:

11.3 Cable installation:

a.Figure 1 shows the area of "J" cable installation. Install the cable according to the numbering label.

b. Install the 4 wire connectors on the MP backplane interface (MPLN_0_J27, MPLN_1_J28, MPLN_2_J27, MPLN_3_J27) according to the labels as shown in the figure.

c. Use 1 cable tie (813-929-00) to fix the leftmost wire on the T/H frame through the slot of the frame as shown in the picture.

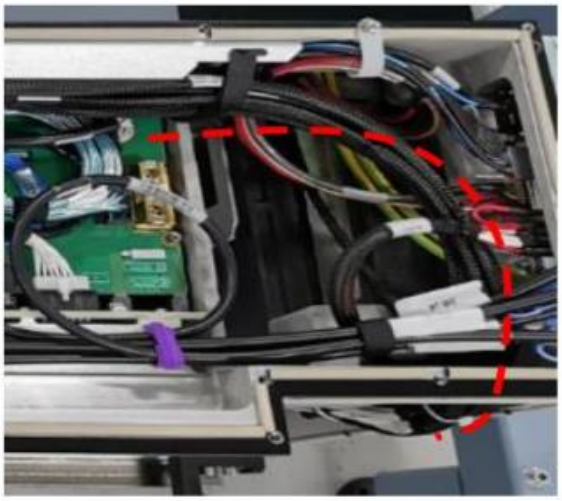
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
32	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE: N/A

METHOD:

12.1 Route the wire through the T/H opening as shown, leaving 14 inches in length and securing with a cable tie (813-929-00)

12.2 Fasten the wire harness with the existing cable ties as shown.